Eta-Quotients in Lean 4

Cusp Orders at Every Cusp, Ligozat's Criterion at Small Level, and a Named Obstruction at General Level

PREPRINT — not peer reviewed

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September 8, 2026

Archived at Zenodo: 10.5281/zenodo.22648098 (all versions)

Status. This is a *preprint*. It has **not been peer reviewed**. The Lean development it describes compiles and is axiom-audited — those claims are machine-checked and independently reproducible — but the *exposition* has had no external referee. One headline result is deliberately bounded (§4): Ligozat's transformation law is proved for $N \in \{1, 2, 3, 4, 5, 7, 13\}$ and **not** in general, the general case being blocked by the obstruction named in §5. This version incorporates revisions from one external review and one internal adversarial pass (the response is archived alongside the deposit); neither constitutes venue peer review. Read the scope before citing the result.

Abstract

We formalize in Lean 4, on top of Mathlib, the arithmetic and analytic layers of Ligozat's criterion for eta-quotients $f(z) = \prod_{\delta|N} \eta(\delta z)^{r_\delta}$, and we report one theorem, one bounded theorem, and one obstruction. **(i)** For *every* level N, every integer exponent vector r with $\sum_{\delta|N} r_\delta = 2k$, and *every* $\gamma \in SL_2(\mathbb{Z})$ — with no congruence condition and no $\Gamma_0(N)$-membership hypothesis — we prove the two-sided asymptotic $\|(f|_k\gamma)(z)\| = \Theta(\exp(-2\pi \operatorname{ord}^{\mathrm{raw}}(N, r, c) \operatorname{Im} z))$ along $\operatorname{Im} z \to \infty$, where $c = \gamma_{10}$ and $\operatorname{ord}^{\mathrm{raw}}(N, r, c) = \frac{1}{24} \sum_{\delta|N} \gcd(c, \delta)^2 r_\delta / \delta$. At the cusp ∞ we additionally identify the order in *Mathlib's own* language, `meromorphicOrderAt (cuspFunction 1 f) 0`. **(ii)** We prove Ligozat's transformation law in full — $f(\gamma z) = w(\gamma)(cz + d)^k f(z)$ on *all* of $\Gamma_0(N)$, with the character identified — for $N \in \{1, 2, 3, 4, 5, 7, 13\}$. **(iii)** For general N we do not prove it, and we explain precisely why: the multiplier is $w(\gamma) = \exp\big(\frac{\pi i}{12} \operatorname{per}(\gamma)\big)$ where per is the period cocycle of the weight-two quasi-modular combination $\sum_\delta r_\delta\, \delta\, E_2(\delta z)$; for a single η this period function is Rademacher's Φ, whose non-coboundary content is the Dedekind sum $s(d, |c|)$. Mathlib contains no Dedekind sums. We state the minimal Mathlib addition that would remove the obstruction. **Postscript (2026-09-08, §8):** a follow-up effort has since built all three items independently in our own library, with attribution to prior art where used; this refutes our own prediction that item (3) alone would make general-N Ligozat "moderate bookkeeping," and narrows the obstruction to one named remaining node rather than removing it, which we report as an addendum, not as a revision of the results proved here. The development is 519 declarations in 8117 lines across five modules, with no `sorry`, pinned by axiom guards in the library's shared `FinalCheck.lean` (797 guards in total as of this printing; see §8 for why that file now also guards material outside this paper's own scope).

MSC 2020: 11F20, 11F11, 68V20. **Keywords:** eta-quotients, Ligozat's criterion, Dedekind eta, Dedekind sums, Rademacher Φ, Lean 4, Mathlib.

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1 Introduction

An *eta-quotient* of level N is a product

$$f(z) \;=\; \prod_{\delta \mid N} \eta(\delta z)^{r_\delta}, \qquad r_\delta \in \mathbb{Z},$$

over the divisors of N, where η is Dedekind's function. Ligozat's criterion states that f is a modular form on $\Gamma_0(N)$ of weight $k = \frac{1}{2}\sum_\delta r_\delta$ provided

$$\sum_{\delta \mid N} \delta\, r_\delta \equiv 0 \pmod{24}, \qquad \sum_{\delta \mid N} \tfrac{N}{\delta}\, r_\delta \equiv 0 \pmod{24}, \tag{1}$$

and the order of f at every cusp is non-negative [1, 2, 3]. It is the standard working tool for producing modular forms by hand, and the reason eta-quotients appear throughout the theory of partitions and of monstrous moonshine.

This paper reports what happens when one tries to formalize it. The honest answer is: the arithmetic and the analysis go through in general; the *group theory* does not, and the obstruction is sharp and classical.

What Mathlib provides, and what it does not

We are explicit about the substrate because it determines what is new. At commit `905b9581` (2026-07-28) Mathlib contains, in `NumberTheory/ModularForms/`:

- `DedekindEta.lean` — 20 declarations: η as the q-product $\eta(z) = q^{1/24}\prod_n (1 - q^{n+1})$, non-vanishing (`eta_ne_zero`), differentiability, and $\mathrm{logDeriv}\,\eta = \frac{\pi i}{12} E_2$;
- `Discriminant.lean` — crucially, the *S-transformation of η*: `eta_comp_eq_csqrt_I_inv` gives $\eta(-1/z) = (\sqrt{i})^{-1}\sqrt{z}\,\eta(z)$, together with the modularity of $\Delta = \eta^{24}$;
- `Cusps.lean` (`IsCusp`, `CuspOrbits`, `widthInfty`), `QExpansion.lean` (`cuspFunction`, `qExpansion`), and the general order machinery `analyticOrderAt` / `meromorphicOrderAt`.

Repository-wide searches return **zero** occurrences of `EtaQuotient`, `Ligozat`, `DedekindSum`, `dedekindSum` and `etaMultiplier`. The last two absences are the subject of §5.

Remark 1.1 (A methodological warning we paid for). An earlier pass in this project concluded that Mathlib had no order-of-vanishing machinery, on the strength of `grep -rl 'orderAt'` returning zero files. It does have it: the declarations are `analyticOrderAt` and `meromorphicOrderAt`, and `grep -rli 'orderat'` returns nineteen. A zero result from a case-sensitive search on a lower-camelCase name is not evidence of absence. Correcting this turned an assumed obstruction into Theorem 3.3.

2 The arithmetic layer

The combinatorial content is separated into a module mentioning no modular form, no η and no upper half-plane: it is a statement about r and N alone. The central definition is Ligozat's cusp-order expression, for $d \mid N$,

$$\mathrm{ord}(N, r, d) \;=\; \frac{N}{24}\sum_{\delta \mid N} \frac{\gcd(d,\delta)^2\, r_\delta}{\gcd\!\big(d, \frac{N}{d}\big)\; d\,\delta},$$

transcribed from [3, Thm. 1.64], together with the two congruences of (1) as predicates `LigozatCongr1` and `LigozatCongr2`. The divisor involution $\delta \mapsto N/\delta$ (Mathlib's `Nat.sum_div_divisors`) gives the duality exchanging the two congruences, formalized as `ligozatCongr1_dual`.

3 Cusp orders, and a correction to our own specification

Theorem 3.1 (Order at every cusp, general N)**.** *etaQuotient_cusp_order. For every N, every r with $\sum_\delta r_\delta = 2k$, and every $\gamma \in SL_2(\mathbb{Z})$, writing $c = \gamma_{10}$,*

$$\big\|(f|_k\gamma)(z)\big\| \;=\; \Theta_{\operatorname{Im} z\to\infty}\; \exp\!\big(-2\pi \operatorname{ord}^{\mathrm{raw}}(N,r,c) \operatorname{Im} z\big), \qquad \operatorname{ord}^{\mathrm{raw}}(N,r,c) = \frac{1}{24}\sum_{\delta|N} \frac{\gcd(c,\delta)^2 r_\delta}{\delta}.$$

```
def etaQuotientRawOrder (N : ℕ) (r : EtaExp) (c : ℤ) : ℚ :=
  (∑ δ ∈ N.divisors, ((Int.gcd c (δ : ℤ) : ℚ) ^ 2 * (r δ : ℚ)) / (δ : ℚ)) / 24

theorem etaQuotient_cusp_order (N : ℕ) (r : EtaExp) {k : ℤ}
    (hk : ∑ δ ∈ N.divisors, r δ = 2 * k) (γ : SL(2, ℤ)) :
    (fun z : ℍ => ‖(etaQuotientH N r ∣[k] ((γ : SL(2, ℤ)) : GL (Fin 2) ℝ)) z‖)
      =Θ[UpperHalfPlane.atImInfty]
      fun z : ℍ => Real.exp (-2 * π * ((etaQuotientRawOrder N r (γ 1 0) : ℚ) : ℝ) * z.im) := by
```

Listing 1: The raw cusp order, and Theorem 3.1. The two-sided bound is Mathlib's `Asymptotics.IsTheta` at the filter `UpperHalfPlane.atImInfty`, written `=Θ[·]`; no approach region appears because that filter constrains only $\operatorname{Im} z$.

Two features deserve emphasis. There is *no* congruence hypothesis and *no* $\Gamma_0(N)$-membership hypothesis: the asymptotic holds for every γ in the full modular group. And the Θ-relation is taken along the filter `atImInfty`, which constrains only $\operatorname{Im} z$; the statement is therefore uniform in $\operatorname{Re} z$ and no approach region is assumed.

Remark 3.2 (The width normalisation — our specification was wrong)**.** This node was originally specified with the exponent written as Ligozat's $\operatorname{ord}(N,r,c)$. *That statement is false*, and it is not what was proved. Ligozat's $\operatorname{ord}(N,r,d)$ is an order of vanishing in the local uniformiser $q_h = e^{2\pi i z/h}$ at a cusp of denominator d, where $h = N/\gcd(d^2,N)$ is the width of that cusp for $\Gamma_0(N)$. A decay statement in $\operatorname{Im} z$ therefore carries the exponent ord/h, not ord. Formally, `etaQuotientCuspOrder_eq_width_mul_raw` gives $\operatorname{ord}(N,r,d) = h\cdot\operatorname{ord}^{\mathrm{raw}}(N,r,d)$, so the two exponentials differ whenever $h \neq 1$. The companion statement `etaQuotient_cusp_order_ligozat` spells the exponent as ord/h so that the correction is visible in the statement itself.

Ligozat's condition (iii) is untouched: $h > 0$, so $\operatorname{ord}^{\mathrm{raw}} \geq 0$ at a cusp if and only if $\operatorname{ord} \geq 0$ there. A hand-computed pin is kept in the development: $f = (\eta(z)\eta(2z))^{12}$ satisfies $\|f|_{12}S\| \asymp \exp(-3\pi y/2)$.

Theorem 3.3 (The cusp ∞, in Mathlib's own language)**.** *meromorphicOrderAt_cuspFunction_eq_cuspOrder_infty. For every N, every r, and every $m \in \mathbb{Z}$ with $24m = \sum_{\delta|N} \delta\, r_\delta$,*

$$\operatorname{meromorphicOrderAt}\big(\operatorname{cuspFunction}_1(f)\big)(0) \;=\; m,$$

and $m = \operatorname{ord}(N,r,N)$, Ligozat's value at $d = N$.

This uses Mathlib's own notion of order, its own cusp function, and its own width (`strictWidthInfty` of $\Gamma_0(N)$ is 1); we introduce no competing notion of "order at a cusp". There is no positivity and no holomorphy hypothesis — the order may be negative.

4 Ligozat's transformation law at small level

Theorem 4.1 (Full transformation law, $N \leq 4$)**.** *ligozat_of_le_four. Let $0 < N \leq 4$, let $\sum_{\delta|N} r_\delta = 2k$, and assume both congruences (1). Let $w : \Gamma_0(N) \to \mathbb{C}^\times$ be any homomorphism* *with* $w(T) = 1$, $w(V) = 1$ *and* $w(-I) = (-1)^k$, *where* $V = W_N T W_N^{-1}$. *Then for every* $\gamma \in \Gamma_0(N)$ *and every* $z \in \mathbb{H}$,

$$f(\gamma z) \;=\; w(\gamma)\,(cz+d)^k\, f(z).$$

The architecture is worth stating because it is what makes the bounded result possible. A general construction produces *a* character with this transformation law for every N and with no congruence conditions; what it cannot do is say *which* character. Ligozat's first congruence pins it at T; the second pins it at $V = W_N T W_N^{-1}$ — and W_N here is the Fricke involution from the companion development [8], which is where that work is reused; a separate argument pins it at $-I$. For $N \le 4$ those three elements generate $\Gamma_0(N)$, by a Euclidean descent, so the character is determined and the congruences are upgraded from "conditions at two generators" to the transformation law on the whole group.

```
noncomputable def etaQuotientModularForm {N : ℕ} (hN : 0 < N) (hN4 : N ≤ 4) (r : EtaExp) {k : ℤ}
    (hk : ∑ δ ∈ N.divisors, r δ = 2 * k) (h1 : LigozatCongr1 N r) (h2 : LigozatCongr2 N r)
    (hke : Even k)
    (hbd : ∀ γ : SL(2, ℤ), IsBoundedAtImInfty ((etaQuotientH N r) ∣[k] γ)) :
    ModularForm (Gamma0 N) k where
  toFun := etaQuotientH N r
  slash_action_eq' := etaQuotientH_slash_of_le_four hN hN4 r hk h1 h2 hke
  holo' := mdiff_etaQuotientH N r
  bdd_at_cusps' hc := isBoundedAt_of_bdd_SL2Z r hbd hc
```

Listing 2: The eta-quotient as a term of Mathlib's bundled `ModularForm`, for $N \le 4$ and even k. Ligozat's condition (iii) enters as the hypothesis `hbd`; see Remark 4.2.

Remark 4.2 (The bundled `ModularForm`, and what condition (iii) costs). For $N \le 4$ and *even* k the eta-quotient is a term of Mathlib's bundled `ModularForm (Gamma0 N) k`: `etaQuotientModularForm`, whose three fields are the transformation law of Theorem 4.1, holomorphy, and boundedness at the cusps. So the assembly is done, not outstanding — but it takes Ligozat's condition (iii) as the hypothesis `hbd`, boundedness of every $SL_2(\mathbb{Z})$-translate at $i\infty$, rather than deriving it from non-negativity of the cusp orders. Deriving it needs the order of vanishing at a *general* cusp, and that is the obstruction of §5 again: the same missing multiplier. Theorem 3.1 supplies the analytic content at every cusp, so what is missing is the bridge from a Θ-asymptotic to `IsBoundedAtImInfty` at a cusp whose width is not 1 — API, resting on an obstructed input.

The even-k hypothesis is not bookkeeping either. For odd k the conclusion is *false*: at $N = 3$, $r_1 = -3$, $r_3 = 9$ both congruences hold and $\sum r_\delta = 6 = 2\cdot 3$, yet the trivial-multiplier conclusion fails at $\gamma = -I$. That counterexample is formalized as a guard.

Theorem 4.3 (Genus-zero primes)**.** *`ligozat_of_prime`. The same conclusion holds for* $p \in \{5, 7, 13\}$.

The descent fails at $N = 5$, and not by accident: the development records the explicit counterexample `descent_bound_fails_at_five`. Classically $\overline{\Gamma}_0(5)$ is the free product $\mathbb{Z}/2 * \mathbb{Z}/2 * \mathbb{Z}$, so $\{-I, T, V\}$ cannot generate it and two elliptic generators are needed. That necessity is a literature fact and is *not* formalized here; what is formalized is that this proof needs them — deleting the two elliptic generators and re-running the block leaves exactly two unprovable goals. The replacement tool is a Schreier transversal criterion, `Gamma0_eq_of_schreier`, valid at every prime p: a subgroup containing $-I$, T, V equals $\Gamma_0(p)$ as soon as it contains one Schreier generator for each $j \in [1, p-1]$. We supply those generators at the three primes where $X_0(p)$ has genus zero.

5 The obstruction at general level

We do not prove Ligozat's transformation law for general N, and the reason is not a shortage of effort or a choice of proof strategy. It is exact.

Write the multiplier as $w(\gamma) = \exp\left(\frac{\pi i}{12}\,\mathrm{per}(\gamma)\right)$, where per is the period cocycle of the weight-two quasi-modular combination $\sum_{\delta|N} r_\delta\, \delta\, E_2(\delta z)$. This period is precisely the integration constant that both natural soft routes destroy: the logDeriv route differentiates it away, and the 24th-power route (passing to f^{24}, a quotient of discriminants whose modularity Mathlib already has) quotients it away. For a single η this period function *is* the classical Dedekind–Rademacher Φ [5, Thm. 3.11]: for $c \neq 0$,

$$\Phi(\gamma) \;=\; \frac{a+d}{c} \;-\; 12\,\mathrm{sign}(c)\, s(d, |c|),$$

whose non-coboundary content is exactly the Dedekind sum $s(d, |c|)$. Its conjugation-invariant homogenization, the Rademacher symbol $\Psi(\gamma) = \Phi(\gamma) - 3\,\mathrm{sign}(c(a+d))$, is what the E_2-period computes over the closed geodesic of a hyperbolic γ [4]; the multiplier itself uses Φ, and it is Φ that obeys the cocycle law quoted below.

The reason is group-theoretic. A character into μ_{24} factors through the abelianization, and for $\Gamma_0(N)$ — a Fuchsian group of genus $g = \mathrm{genus}(X_0(N))$ with s cusps — the free rank of the abelianization is $2g+s-1$ (a classical fact, not formalized here). The congruences pin w only at T and at $V = W_N T W_N^{-1}$, hence on a subgroup of the free part of rank at most two, and torsion relations pin it at elliptic points; on a generator of infinite order, a relation such as $w^{24} = 1$ says only that the image lies in μ_{24}, never *which* root of unity it is. The small-level successes of §4 are exactly the levels where this bookkeeping closes: for $N \leq 4$ the free rank is at most two, and at $p \in \{5, 7, 13\}$ it is one, the elliptic pins supplying the rest. The first failure is parabolic, not hyperbolic: $\Gamma_0(6)$ is torsion-free of free rank three (genus zero, four cusps), so $\{-I, T, V\}$ cannot generate it, and the extra parabolic classes could only be pinned through q-expansions at the extra cusps — precisely the general-cusp machinery this section shows to be obstructed. From genus one on (for instance $N = 11$, and most $N \geq 17$) the situation is strictly worse: $2g$ independent homology classes are represented by *hyperbolic* elements, on which no congruence, torsion relation, or cusp expansion says anything, and pinning w there requires evaluating the period integral explicitly — which is what forces Dedekind sums into the argument. Evaluating w in closed form on a single hyperbolic element of $\Gamma_0(11)$ is mathematically equivalent to evaluating a Dedekind sum. Every disguise we attempted reduces back to this: Atkin–Lehner conjugation reintroduces general eta transforms, theta/Poisson routes re-derive the same 24th-root phases, and Wohlfahrt level-24 arguments presuppose the multiplier formula.

What would remove it

In ascending strength, the minimal additions are:

(1) the Dedekind sum $s(d, c)$ and reciprocity;

(2) Rademacher's $\Phi : SL_2(\mathbb{Z}) \to \mathbb{Z}$ with the cocycle law $\Phi(AB) = \Phi(A) + \Phi(B) - 3\,\mathrm{sign}(c_A c_B c_{AB})$;

(3) the Petersson–Rademacher multiplier $\eta(\gamma z) = \exp\left(\pi i \left(\frac{a+d}{12c} - s(d, c) - \frac{1}{4}\right)\right) \sqrt{cz+d}\;\, \eta(z)$ for $c > 0$.

This list, and what follows from it, is corrected in §8. A first draft of this paper claimed all three already existed, sorry-free, in the Fermat's Last Theorem artifact [7], and predicted that item (3) alone would reduce general-N Ligozat to "moderate bookkeeping." Both claims were checked by attempting the construction, and both were found wrong: item (2) does not exist upstream as a named function with the cocycle law above — only a single Euclidean-descent *step* does, from which Φ can be built but is not built there — and item (3), once actually in hand, exposed a further genuine theorem rather than bookkeeping. See §8 for the corrected account, including what has since been proved.

Remark 5.1 (A per-instance escape hatch). For a *fixed* N and r, the multiplier on a finite generating set is a ratio of convergent q-products valued in the discrete set μ_{24}. Interval arithmetic with error below $|1 - \zeta_{24}|/2$ therefore *proves* each value. This is an *A*2 (program-checked) route for any concrete eta-quotient at any level, requiring no Mathlib addition, though heavy to promote to kernel-checked *A*1.

6 Verification status

`lake build SocrateAI` — the whole library's default target, now shared with the postscript's material — completes in 3767 jobs with no errors. This paper's own development is 519 declarations in 8117 lines across five modules (`EtaQuotient`, `EtaQuotientCuspOrder`, `EtaQuotientCuspTheta`, `EtaQuotientModularity`, `EtaQuotientPrimeLevel`). There are no `sorry` placeholders, and the five eta modules declare no axioms. (One legacy module elsewhere in the library, `Generated/BlueprintSkeleton.lean`, declares six vacuous `axiom ... : True` postulates from an early scaffold; they are imported nowhere, and the enforced footprints below prove that no guarded theorem depends on them. An earlier version of this sentence said five — the count came from a truncated listing, and is now recomputed mechanically at every build of this paper.) Following [7], the axiom footprint is enforced rather than reported: `FinalCheck.lean` carries 797 `#guard_msgs in #print axioms` guards over 795 distinct theorems (the library-wide total; §8 accounts for the material outside this paper's own scope), so a widened footprint is a compile error; 769 report exactly [`propext`, `Classical.choice`, `Quot.sound`], two the strictly smaller [`propext`, `Quot.sound`], sixteen (definitional guards) depend on no axiom at all, and eight are *inverted tripwires* — guards whose EXPECTED footprint includes `sorryAx`, asserting that a specific, DAG-tracked open or refuted statement still depends on one, so that a later accidental proof of it (or a silent weakening that lets it typecheck) fails the build rather than passing unnoticed (§8). A negative control asserting a deliberately wrong footprint is kept outside the build and verified to fail, which is what establishes the guards are load-bearing rather than vacuous.

Of the 453 theorems in the new modules, 14 (3%) have a one-line `rfl/decide` proof, and 45% quantify over a structure. We report this ratio because an earlier module in this project consisted of 174 numeral identities of which 158 were one-line, and carried no evidential weight; the check is now standard practice here for any large batch.

The statement-dependency graph (`dag/theorems.jsonl`, 74 nodes, 66 proved, acyclic — library-wide, per §8) is machine-validated: every node marked proved must name a declaration that resolves and may depend only on proved nodes. **The node `ETA-01`, Ligozat's criterion in general, remains `open` with `lean_name: null`**, and the obstruction node is `blocked` with `lean_name: null` — for that node the absence of a declaration *is* the content.

Artifact. Repository https://github.com/xaviercallens/SocrateAI-Lean-Lib; the five modules `EtaQuotient*.lean` under `Lean/SocrateAI/ModularForms/`, axiom guards in `Lean/SocrateAI/FinalCheck.lean`, dependency graph in `dag/`. Toolchain `leanprover/lean4:v4.32.2`; Mathlib pinned at `905b95818eb32af7874a58b427f50c1711a5e96c` (2026-07-28), which is the revision every theorem here was checked against. The build configuration is portable — `lakefile.lean` requires Mathlib from git at that revision and no tracked file contains a machine-specific path — so `git clone`, `lake exe cache get`, `lake build SocrateAI` reproduces the artifact; see `BUILDING.md`. Cite this preprint through its concept DOI [11], which resolves to the newest version. Mirrored at https://huggingface.co/datasets/callensxavier/socrateai-eta-quotients.

7 Related work and prior art

Mathlib has no eta-quotients. The Fermat's Last Theorem artifact [7] contains substantial eta-related material of two distinct kinds, and an earlier version of this paper described only the first. **(a)** Modular *units* on modular function fields — `modularUnitSeries`, `sharpUnit*`, over Laurent and Hahn series; its three occurrences of "Ligozat" are *namespace labels* (`LigozatUnitEngine`, `LigozatUnitAL`) on that machinery, not a formalization of the modularity criterion. That reading stands. **(b)** *The arithmetic of the eta multiplier itself* — which we missed, and which is exactly what §5 identifies as missing from Mathlib. The artifact defines the sawtooth and the Dedekind sum (`dedekindSaw`, `dedekindSum`), proves **reciprocity** (`dedekindSum_add_dedekindSum`: $s(h,k) + s(k,h) = (h/k + k/h + 1/(hk))/12 - 1/4$ for coprime positive h, k), proves the Rademacher Φ descent step (`rademacher_phi_step`, with $\Phi(\gamma) = (a+d)/c - 12\,s(d,c)$), and — decisively — proves the full Petersson–Rademacher transformation

$$\eta(\gamma z) \;=\; \exp\!\Big(\tfrac{\pi i}{12}\big(\tfrac{a+d}{c} - 12\,s(d,c)\big)\Big)\,\sqrt{-i(cz+d)}\;\eta(z), \qquad c > 0,$$

as `ModularForm.eta_specialLinearGroup_smul`. All are `sorry`-free. This is items (1), (2) and (3) of §5 — the entire list — already formalized in Lean 4. A GitHub-wide `language:lean` search returns zero for `Ligozat`. We did *not* search the Lean Zulip or open mathlib4 pull requests, and we do not claim coverage there.

Other proof assistants. The component we identify as obstructed has been mechanized elsewhere, which is evidence that the obstruction is a gap in Mathlib rather than a difficulty of principle. The Isabelle/HOL Archive of Formal Proofs contains *Dedekind Sums* [9], which defines $s(h,k) = \sum_{r=1}^{k-1} \frac{r}{k}\left(\left\{\frac{hr}{k}\right\} - \frac{1}{2}\right)$ and proves reciprocity, and *Rademacher's Series for the Partition Function* [10], which uses that machinery in exactly the analytic setting where it arises here. Our §5 therefore asks Mathlib for something Isabelle/HOL already has; the constructions there are a usable guide, and the existence of two independent formalizations of Dedekind sums is a reason to expect the Mathlib addition to be routine rather than research.

8 Postscript (2026-09-08): the obstruction, attempted

Section 5's "what would remove it" list was, in an earlier version of this paper, presented as already available upstream. A dedicated follow-up run attempted the construction directly: a first, unscoped attempt was correctly *aborted at its own prior-art gate* when it found genuine Lean 4 material for all three items in the Fermat's Last Theorem artifact [7] — but that finding was itself imprecise, as the second pass below corrects. A second, narrower run then ported and independently built the three items in this library, with attribution recorded declaration by declaration in the artifact's `ATTRIBUTION.md`.

The corrected account of item (2). FLT does *not* contain Rademacher's Φ as a named function with the cocycle law stated in §5. It contains one Euclidean- descent *step* relating Φ at a matrix to Φ at a smaller one, which is not the same statement and does not by itself give the function. Our own `rademacherPhi` is defined directly from Apostol's closed form and its cocycle law is proved from our own reciprocity theorem, independently. This is item (2) genuinely built, not ported — FLT gets us only partway there.

What is now proved, sorry-free, in this library. The Dedekind sum, its sawtooth, and nineteen supporting lemmas are a verbatim **port** of FLT's `Definitions/Def_NumberTheory_DedekindSum.lean` (Apache-2.0; attributed). Reciprocity, the Φ-descent step, and the Petersson–Rademacher multiplier itself are **statement-only ports** — the theorem signatures match FLT's exactly, so a reader can compare term for term — with proofs built independently against this library's own architecture, reusing the Fricke involution W_N from the companion development [8] for the multiplier's Euclidean-descent induction. None of FLT's proof text was available to the agent that wrote these proofs: it was fetched separately, for an unrelated `sorry`-count check, in a different step of the run.

The prediction was wrong, and here is why. Item (3) in hand does *not* reduce general-N Ligozat to bookkeeping. What it buys is exactly the analytic layer: the multiplier becomes a closed-form exponential at every γ with $c > 0$, and the criterion becomes *equivalent* to a single congruence mod 24 in a quantity we call $\Phi_N(r, \gamma) = \sum_{\delta|N} r_\delta \, \Phi(\gamma_\delta)$ over the divisor-conjugated matrices. What remains is a genuine arithmetic theorem: a divisor-by-divisor quadratic-reciprocity assembly turning the numerator of one Dedekind sum into Ligozat's numerator δ, together with an even-modulus companion to our own odd-weight Jacobi-symbol evaluation of $12c\, s(d, c)$. That theorem is not proved here; it is the one node, precisely named, that separates the present state from a closed proof of Ligozat's criterion at every level.

What was proved instead. The $c \le 0$ strata are closed *at every* N: the multiplier's behaviour under $\gamma \mapsto -\gamma$ and the Kronecker character's behaviour under $d \mapsto -d$ both contribute the same sign, and they cancel. Combined with Theorem 4.1, Ligozat's criterion is now proved in full, ending in a genuine `ModularForm (Gamma0 N) k` term, for $0 < N \le 4$. And a machine-checked **refutation** was found along the way: an earlier, easier-looking general-N statement in this library asserted that Ligozat's congruences together with $12 \mid k$ force a *trivial* multiplier on all of $\Gamma_0(N)$. That statement is **false**. At $N = 17$, $r = (r_1, r_{17}) = (21, 3)$, $k = 12$, $\gamma = \left(\begin{smallmatrix} 6 & 1 \\ 17 & 3 \end{smallmatrix}\right) \in \Gamma_0(17)$, every hypothesis holds and the multiplier is -1, not 1 — confirmed independently by a direct 4000-term η-product evaluation and by three other routes agreeing to high precision. The false statement has been renamed, quarantined (its proof obligation is kept as a permanent, disclosed `sorry` rather than deleted, so that the refutation has a fixed target), and removed from the library's default build target. That an easy generalization failed here is itself evidence for why the correct one — via the Kronecker character, not a divisibility hypothesis — is the theorem this paper's Theorem 4.1 and Theorem 4.3 actually prove.

Net effect on the DAG. Before this postscript, node `ETA-01` (Ligozat's criterion at general N) had three missing inputs. It now has one: the reciprocity-assembly theorem named above. Two new results close cleanly around it: the full $c \le 0$ reduction at every N, and Ligozat's criterion in full for $0 < N \le 4$ as a genuine bundled modular form. The node itself remains `open` with `lean\_name: null`; we do not claim it, here or anywhere in this paper.

On self-review. Every technical claim in this postscript — the $c \le 0$ sign cancellation, the Φ_N mod-24 equivalence, and the $N = 17$ witness — was checked against the artifact's own statement-dependency graph and, for the witness, against an independent numerical evaluation, before this text was allowed to stand; none was taken from a subagent's summary alone. Text written to correct an error gets the same scrutiny as the error it corrects (this is a standing rule of the programme after an earlier correction pass introduced two errors of its own while fixing others).

9 Limitations

Remark 9.1 (What is not done).

(a) Ligozat's criterion for general N is not proved (§5).

(b) The order of vanishing at a cusp *other than* ∞ is not stated in Mathlib's `meromorphicOrderAt` language. Mathlib's width is `widthInfty` only; there is no `widthAtCusp` and no `orderAtCusp`. These are constructible from `conjGL`, `widthInfty` and the slash action, so this is missing API rather than an obstruction. Theorem 3.1 gives the analytic content at every cusp regardless.

(c) The indexing data is not formalized: that the cusps of $\Gamma_0(N)$ are parametrised by $d \mid N$ with multiplicity $\varphi(\gcd(d, N/d))$, and that $[SL_2(\mathbb{Z}) : \Gamma_0(N)] = \psi(N)$. No index formula for Γ_0 exists anywhere in Mathlib. The definitions `numCusps` and `dedekindPsi` in this development carry those classical readings but do not prove them.

(d) Ligozat's condition (iii) is not *derived* from the cusp-order formula. The bundled `ModularForm` exists (Remark 4.2) but takes boundedness as a hypothesis; discharging it from non-negativity of the cusp orders needs the order at a general cusp, which is (b), which is the obstruction.

(e) That $\{-I, T, V\}$ *cannot* generate $\Gamma_0(5)$ is cited, not proved.

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